

Homework (Habanero)

Q1. Rationalize the denominator of $\frac{2}{\sqrt{3}}$

- (A) $\frac{2\sqrt{3}}{3}$
- (B) $\frac{2}{3\sqrt{3}}$
- (C) $\frac{\sqrt{3}}{2}$
- (D) $\frac{2}{3}$
- (E) $\sqrt{3}$

Q2. Simplify $\frac{1}{\sqrt{7}+1}$ by rationalizing the denominator.

- (A) $\frac{1}{\sqrt{7}-1}$
- (B) $\frac{\sqrt{7}-1}{6}$
- (C) $\frac{\sqrt{7}+1}{6}$
- (D) $\sqrt{7}-1$
- (E) $\sqrt{7}+1$

Q3. Rationalize the denominator of $\frac{3}{2\sqrt{5}-1}$.

- (A) $\frac{3(2\sqrt{5}+1)}{19}$
- (B) $\frac{3(2\sqrt{5}-1)}{20}$
- (C) $\frac{2\sqrt{5}+1}{3}$
- (D) $\frac{2\sqrt{5}-1}{3}$
- (E) $\frac{3}{2\sqrt{5}}$

Q4. The volume of a cell is given by $V = \frac{4}{3}\pi r^3$, where r is the radius of the cell in $\sqrt{\mu m}$. If $r = 2\sqrt{3}$, find the volume.

- (A) $\frac{32\pi}{\sqrt{3}}$
- (B) $\frac{32\pi\sqrt{3}}{3}$
- (C) $\frac{32\pi\sqrt{3}}{9}$
- (D) 32π
- (E) $\frac{32\pi}{3}$

Q5. A medicine's effectiveness is given by $\frac{100}{\sqrt{t}}$

- (A) $\frac{100\sqrt{9}}{9}$
- (B) $\frac{100}{3}$
- (C) $\frac{100\sqrt{3}}{3}$
- (D) 33.33
- (E) 11.11

Q6. In a healthcare study, the average height of patients is $\frac{175}{\sqrt{3}}$ cm. Rationalize the denominator.

- (A) $\frac{175\sqrt{3}}{3}$ cm
- (B) 58.33 cm
- (C) $\frac{175}{3}$ cm
- (D) $\frac{175\sqrt{2}}{2}$ cm
- (E) $\frac{175}{2}$ cm

Q7. Simplify $\frac{2+\sqrt{3}}{2-\sqrt{3}}$ by rationalizing the denominator.

- (A) $\frac{4+2\sqrt{3}}{1}$
- (B) $\frac{2+\sqrt{3}}{2-\sqrt{3}}$
- (C) $\frac{2-\sqrt{3}}{2+\sqrt{3}}$
- (D) $2 + \sqrt{3}$
- (E) $\frac{-4+2\sqrt{3}}{1}$

Q8. A biologist is studying cell growth, which can be modeled by $\frac{dN}{dt} = \frac{1}{\sqrt{t}+2}$. Rationalize the denominator of $\frac{1}{\sqrt{t}+2}$.

- (A) $\frac{\sqrt{t}-2}{t-4}$
- (B) $\frac{\sqrt{t}+2}{t+4}$
- (C) $\frac{\sqrt{t}-2}{t+4}$
- (D) $\frac{1}{\sqrt{t}-2}$
- (E) $\sqrt{t}+2$

Open Ended Questions (FRQ)

Q9. Rationalize the denominator of $\frac{1}{\sqrt{x}+1}$ and then simplify.

Q10. The dosage of a medicine is given by $\frac{2}{\sqrt{d}}$ mg, where d is the distance from the injection site in $\sqrt{cm}$. If $d = 16$, find the dosage and rationalize the denominator.

Chef's Tips

- Use a calculator only when necessary.
- Show all steps for free response questions.
- Check your answers.